ASESII 24A02 Final Project - Part 2

2-way ANOVA method

Group 02
Nguyen Thi Yen Nhi (24125071)
Nguyen Khang Thinh (24125104)
Nguyen Van Tinh (24125106)
Nguyen Le Quoc Huy (24125100)

2026-08-19

Contents

1 Abstract

Reproducible setup

```
required_packages <- c("tidyverse", "glmnet", "broom", "knitr", "car", "ggplot2")
missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]
if (length(missing_packages)) {
  stop("Install the required packages before rendering: ",
       paste(missing_packages, collapse = ", "))
}

library(tidyverse)
library(glmnet)
library(broom)
```

```r
library(knitr)
library(car)
library(ggplot2)
# `params` exists only while knitting; the fallback
# keeps interactive chunk runs working.
has_params <- exists("params") && !is.null(params$group_number)
group_number <- if (has_params) as.integer(params$group_number) else 4L
if (length(group_number) != 1L || is.na(group_number) || group_number < 1L) {
  stop("Set params$group_number to your assigned positive group number.")
}

seeds <- list(
  split = 240200L + group_number,
  cv = 240300L + group_number,
  features = 240400L + group_number
)
```

Shared helper functions

```r
find_exam_data <- function(filename) {
  input <- knitr::current_input(dir = TRUE)
  input_dir <- if (length(input) && nzchar(input)) dirname(input) else getwd()
  candidates <- c(
    file.path(input_dir, "data", filename),
    file.path(input_dir, "..", "data", filename),
    file.path(getwd(), "data", filename)
  )
  existing <- candidates[file.exists(candidates)]
  if (!length(existing)) stop("Cannot locate ", filename, " by a relative path.")
  hits <- unique(normalizePath(existing, winslash = "/", mustWork = TRUE))
  if (length(hits) > 1L && length(unique(unname(tools::md5sum(hits)))) > 1L) {
    stop("Multiple nonidentical copies of ", filename, " were found.")
  }
  hits[[1L]]
}

split_rows <- function(n, train_prop = 0.80, seed) {
  stopifnot(n >= 10L, train_prop > 0, train_prop < 1)
  set.seed(seed)
  train <- sort(sample.int(n, floor(train_prop * n), replace = FALSE))
  list(train = train, test = setdiff(seq_len(n), train))
}

fit_scaler <- function(x, tol = 1e-12) {
  x <- as.matrix(x)
  center <- colMeans(x)
```

```r
  centered <- sweep(x, 2L, center, "-")
  scale <- sqrt(colMeans(centered^2))
  keep <- is.finite(scale) & scale > tol
  if (!any(keep)) stop("No nonconstant training predictor remains.")
  list(
    center = center[keep],
    scale = scale[keep],
    keep = keep,
    dropped = colnames(x)[!keep]
  )
}

apply_scaler <- function(x, prep) {
  x <- as.matrix(x)[, prep$keep, drop = FALSE]
  x <- sweep(x, 2L, prep$center, "-")
  sweep(x, 2L, prep$scale, "/")
}

make_foldid <- function(n, k = 5L, seed) {
  if (k < 3L || k > n) stop("Require 3 <= k <= number of training rows.")
  set.seed(seed)
  sample(rep(seq_len(k), length.out = n))
}

fit_cv_glmnet <- function(x, y, alpha, foldid) {
  glmnet::cv.glmnet(
    x = x,
    y = y,
    family = "gaussian",
    alpha = alpha,
    foldid = foldid,
    standardize = FALSE,
    intercept = TRUE,
    type.measure = "mse"
  )
}

score_regression <- function(y, prediction) {
  y <- as.numeric(y)
  prediction <- as.numeric(prediction)
  if (length(y) != length(prediction) || any(!is.finite(c(y, prediction)))) {
    stop("Metrics require equal-length finite vectors.")
  }
  c(RMSE = sqrt(mean((y - prediction)^2)),
    MAE = mean(abs(y - prediction)))
}

count_nonzero <- function(fit, s = "lambda.min", tol = 1e-8) {
```

```r
  b <- as.matrix(stats::coef(fit, s = s))
  slopes <- b[rownames(b) != "(Intercept)", 1L]
  sum(abs(slopes) > tol)
}

safe_condition_numbers <- function(x, lambda, tol = 1e-10) {
  eigenvalues <- eigen(
    crossprod(x) / nrow(x),
    symmetric = TRUE,
    only.values = TRUE
  )$values
  largest <- max(eigenvalues)
  smallest <- max(min(eigenvalues), 0)
  cutoff <- tol * max(1, largest)
  c(
    gram = if (smallest <= cutoff) Inf else largest / smallest,
    ridge = (largest + lambda) / (smallest + lambda)
  )
}
analysis_model <- function(x, y, alpha, foldid) {
  model <- fit_cv_glmnet(x = x, y = y, alpha=alpha, foldid = foldid)
  list(
    model = model,
    lambda_min = model$lambda.min,
    mse_min = model$cvm[model$lambda == model$lambda.min],
    cond_min = safe_condition_numbers(x, lambda = model$lambda.min)[2],
    coef_min = coef(model, "lambda.min"),
    coef_min_norm = sqrt(sum(coef(model, "lambda.min")[-1]^2)),
    nonzero_min = count_nonzero(model, s="lambda.min"),
    lambda_1se = model$lambda.1se,
    cond_1se = safe_condition_numbers(x, lambda = model$lambda.1se)[2],
    mse_1se = model$cvm[model$lambda == model$lambda.1se],
    coef_1se = coef(model, "lambda.1se"),
    coef_1se_norm = sqrt(sum(coef(model, "lambda.1se")[-1]^2)),
    nonzero_1se = count_nonzero(model, s="lambda.1se")
  )
}
```

2 Dataset selection

```r
data <- read.csv("data/student_performance.csv", header = FALSE)
colnames(data) <- c(
  "student_id",
  "gender",
  "study_time_hours",
```

```
  "attendance_percent",
  "sleep_hours",
  "parental_education",
  "internet_access",
  "extracurricular_activities",
  "part_time_job",
  "previous_grade",
  "final_exam_score",
  "final_grade"
)

config <- list(
  file = "student_performance.csv",
  response = "final_exam_score",
  excluded = "final_exam_score"
)
data_raw <- read.csv(find_exam_data(config$file), check.names = FALSE)
predictors <- setdiff(names(data_raw), config$excluded)
analysis_data <- data_raw[, c(config$response, predictors), drop = FALSE]

if (anyNA(analysis_data)) stop("Declare a training-only missing-value plan.")
```

Table 1: Response and candidate predictors, with the integrity checks run before the split.

Variable	Role	Type	Distinct	Missing
student_id	Predictor	integer	1000	0
gender	Predictor	character	2	0
study_time_hours	Predictor	numeric	70	0
attendance_percent	Predictor	numeric	327	0
sleep_hours	Predictor	numeric	65	0
parental_education	Predictor	character	5	0
internet_access	Predictor	character	2	0
extracurricular_activities	Predictor	character	2	0
part_time_job	Predictor	character	2	0
previous_grade	Predictor	numeric	434	0
final_exam_score	Response	numeric	342	0
final_grade	Predictor	character	5	0

For 2-Way ANOVA, this method requires the dataset to have at least 2 categorical variables along with a continuous response variable for the analysis.

In general, from our chosen dataset, we decide to evaluate the factors that directly affect the student performance on the score. This data constains information over 1000 students, including the daily habits (the study time, hours of sleep, internet access), lifestyles (extracurricular activities, parttime job) demographic information (gender, education level of their parents) and academic performance (final exam score, final grade). It's ideal for exploratory data analysis and 2-Way ANOVA analysis based on the behavioral and demographic factors.

Specifically, there are 6 categorical factors: `gender` (Male/Female), `parental_education`(PhD/Master/Bachelors/School/None), `internet_access`(Yes/No), `extracurricular_activities`(Yes/No), `part_time_job`(Yes/No), `final_grade`(A/B/C/D/F). In this method, we pick the continuous response on `final_exam_score` to test the hypothesis whether any factors that affect this response as well as any relationships/interactions between 2 categorical factors.

2.1 Research Questions & Hypotheses

We investigate whether extracurricular activities and part-time jobs affect students' final exam scores, and whether there is an interaction between the two factors

We follow the 2-way ANOVA model:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

with $\epsilon_{ijk} \sim N(0, \sigma^2)$.

- μ is the overall mean.
- α_i is the effect of the i-th level of Factor A, `extracurricular_activities`.
- β_j is the effect of the j-th level of Factor B, `part_time_job`.
- $(\alpha\beta)_{ij}$ is the interaction effect between Factor A and Factor B.
- ϵ_{ijk} is the random error term.

2.1.1 Factor A: Extracurricular Activities

Research Question:
Does participation in extracurricular activities affect students' final exam scores?

$$\begin{cases} H_{0a} : \alpha_1 = \alpha_2 = 0 \\ H_{1a} : \text{Not all } \alpha_i \text{ are equal to 0.} \end{cases}$$

2.1.2 Factor B: Part-time Job

Research Question:
Does having a part-time job affect students' final exam scores?

$$\begin{cases} H_{0b} : \beta_1 = \beta_2 = 0 \\ H_{1b} : \text{Not all } \beta_j \text{ are equal to 0.} \end{cases}$$

2.1.3 Interaction Effect

Research Question:
Does the effect of extracurricular activities on final exam scores depend on whether students have a part-time job?

$$\begin{cases} H_{0ab}: (\alpha\beta)_{ij} = 0 \quad \text{for all } i, j \\ H_{1ab}: \text{Not all } (\alpha\beta)_{ij} \text{ are equal to 0.} \end{cases}$$